
DS2020– Artificial Intelligence

Lab 5

Due on 23/4/2026 11.59pm

Instructions: Upload to your moodle account one zip file containing the following. Late submission is not allowed without prior approval of the instructor. You are expected to follow the honor code of the course while doing this homework. This lab may be completed in pairs.

1. Name the zip file rollnum1_rollnum2.zip
 2. The zip must contain: l5.py, answers.json, plots and all .pkl files produced by the script
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Classification using Decision Trees

In this lab, you will implement a **Decision Tree classifier** using the **scikit-learn** library to predict the survival of passengers who travelled in the Titanic. The dataset comprises of *1309* observations and *10* features.

Feature	Description	Type
pclass	Ticket class: 1 = Upper, 2 = Middle, 3 = Lower	integer
gender	Gender of the passenger: 0 = male, 1 = female	binary
age	Age of the passenger in years	float
sibsp	Number of siblings or spouses aboard	integer
parch	Number of parents or children aboard	integer
fare	Fare paid by the passenger for the ticket	float
embarked_C	Embarked from Cherbourg port: 1 = yes, 0 = no	binary
embarked_Q	Embarked from Queenstown port: 1 = yes, 0 = no	binary
embarked_S	Embarked from Southampton port: 1 = yes, 0 = no	binary
survived	TARGET: survived (1) or not (0)	binary

1. Reading the Data (5)

We will use the pandas library to read the dataset and create a dataframe. Load the csv file into a pandas DataFrame. Separate the data into features X (all columns except survived) and target labels y (the survived column).

<https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.html>

2. Creating the Train-Test Split (5)

Use the `train_test_split` function from `sklearn.model_selection` to divide the dataset into training and test sets with `test_size` set as 0.2.

https://scikit-learn.org/0.19/modules/generated/sklearn.model_selection.train_test_split.html

3. Train a Decision Tree Classifier (15)

Fit a `DecisionTreeClassifier` on the training set with the default parameter settings given in the starter file. Use the trained model to obtain predictions on `X_test`.

<https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeClassifier.html>

Using the trained classifier, print:

- The mean accuracy on the training set and test set (use `.score()`)
- The depth of the classifier tree
- The two most important features (use `feature_importances_`)

Save the trained classifier as `decision_tree_baseline.pkl`.

4. Decision Tree with `max_depth=3` (15)

Fit a `DecisionTreeClassifier` on the training set with `max_depth` set as 3.

- Print the mean accuracy on the training set and test set
- Compare the mean accuracy on the training and test sets with those for the previous model
- Visualize the tree using `plot_tree()`. Save the figure as `tree_depth3.png`.
https://scikit-learn.org/stable/modules/generated/sklearn.tree.plot_tree.html
- From the plot, identify the most important feature used by the decision tree. How did you identify it?
- Print the most important feature using `feature_importances_` and verify your answer to (d).

Save the trained classifier as `decision_tree_depth3.pkl`.

5. Add Noise to the Training Dataset (10)

Add noise to the training set by randomly flipping the labels with $p_{\text{Flip}} = 0.1$.

Compute the function `flip` that takes a single label (0 or 1) as argument and performs a flip with probability p_{Flip} . Generate a random value uniformly distributed between 0 and 1. If the generated value is less than p_{Flip} , flip the label (convert 0 to 1 and vice versa).

6. Fit a Decision Tree Classifier on the Noisy Dataset (10)

Using the noisy labels `yn_train` from Task 5, fit a `DecisionTreeClassifier` with default settings given in the starter code. Print:

- The mean accuracy on the training set and test set
- The depth of the classifier tree
- The two most important features

Save the trained classifier as `clf_noisy.pkl`

7. Observations (15)

Repeat Tasks 5 and 6 at least 5 times (using different random seeds: 0, 1, 2, 3, 4) and write down your observations about the following.

- How does the average accuracy on the training and test datasets vary before and after noise to the training set?
- Does the average depth of the classifier tree change before and after adding noise to the training set? Explain
- Do the two most important features of the decision tree remain the same before and after adding noise to the training set?

8. Train and Test Accuracy Plot (10)

Fit a `DecisionTreeClassifier` for different values of `max_depth`. Make a single line plot that shows the training and test accuracies for each value of `max_depth` in the range (3, 10). Save the figure as `accuracy_vs_depth.png`.

Use `matplotlib.pyplot.plot()` to generate the line plot.

https://matplotlib.org/3.5.3/api/as_gen/matplotlib.pyplot.html

9. Observations about plot from Task 8 (15)

Open the `accuracy_vs_depth.png` you generated in Task 8. Study the two curves carefully. Answer each question below in `answers.json` under the “q9” key.

Each question has two fields: “answer” (what you read from the plot) and “reasoning” (one sentence explaining why).

Q9.1 Describe the relationship you observe between `max_depth`, training accuracy and test accuracy in your plot. Name the phenomenon. Explain why this behavior occurs in decision trees

Q9.2 Identify the value of `max_depth` where your test accuracy is highest. Report the corresponding test accuracy value from your plot. State which depth you would choose for deployment and justify your choice based on your observed trade-off.

Q9.3 Compute the gap (train accuracy – test accuracy) at:

- the smallest depth in your plot
- the largest depth in your plot

Compare these gaps: Does the gap increase, decrease or stay similar

Explain what this indicates about model generalization as depth increases